

CONTACT DETAILS	Email: ghoshadi@stanford.edu Address: Computing and Data Science (CoDa) Webpage: ghoshadi.github.io 389 Jane Stanford Way, Stanford University Stanford, CA 94305
EDUCATION	Ph.D. in Statistics, Stanford University GPA 4.13/4.00 2022 – 2027 (expected) - Advised by Stefan Wager (Stanford GSB) & Dominik Rothenhäusler (Stanford Statistics) Masters of Statistics (M.Stat), Indian Statistical Institute , Kolkata 2020 – 2022 Dissertation advisor: Prof. Bodhisattva Sen (Columbia University) Bachelor of Statistics (B.Stat), Indian Statistical Institute , Kolkata 2017 – 2020
RESEARCH	My research focuses on problems in Causal Inference, Statistical Learning, and Optimization. I am currently working on Causal Inference in Dynamic Systems (with Stefan Wager) and Causal Inference under Distribution Shift (with Dominik Rothenhäusler). Ghosh, A. & Wager, S. (2025). Non-parametric Causal Inference in Dynamic Thresholding Designs. <i>Submitted</i>. arXiv ↗ <i>Extended abstract accepted at EC 2026</i> (Summary: We study thresholding designs in general dynamic systems, and show that simple reduced-form characterizations remain available for a relevant causal target, namely a dynamic marginal policy effect at the treatment threshold.) Ghosh, A. & Rothenhäusler, D. (2025). Which Covariates to Adjust for? Specification-robust Causal Inference in Observational Studies. <i>Submitted</i>. arXiv ↗ (Summary: When it is unclear which covariates to adjust for, we provide valid inference for a reweighted population when at least one of the candidate adjustment sets is valid.) Ghosh, A., Imbens, G. & Wager, S. (2025). PLRD: Partially Linear Regression Discontinuity Inference. <i>Submitted</i>. arXiv ↗ R package ↗ (Summary: For treatment rules with an eligibility threshold, PLRD delivers narrow yet reliable confidence intervals for the effect at the threshold by solving a minimax problem.) Ghosh, A., Deb, N., Karmakar, B., & Sen, B. (2025+). Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A design-based perspective. 'Accepted' at <i>Biometrika</i>. arXiv ↗ Journal link ↗ (Summary: We study efficiency and robustness of Rosenbaum's rank-based estimator under finite population inference in randomized trials, with/without covariate adjustment.) Ghosh, A. (2019). An asymptotic formula for the Chebyshev theta function. <i>Notes on Number Theory and Discrete Mathematics</i>, 25(4), 1-7. Journal link ↗ (Summary: We derive sharp bounds on the geometric mean of the first n prime numbers.)
ACADEMIC SERVICES	- I help organize the Online Causal Inference Seminar, a weekly seminar on causal inference with participants from all over the world. - Served as a reviewer for: <i>AoS</i>, <i>JASA</i>, <i>JRSSB</i>, <i>NeurIPS</i>
TEACHING	As instructor, Stanford University ExploreCourses ↗ - Stats 302: Qualifying Exam Workshop (Theoretical Statistics). Summer 2025, 2026 - Stats 302: Qualifying Exam Workshop (Probability). Summer 2024

As teaching assistant, **Stanford University** [ExploreCourses](#) [↗](#)

- Stats 361: Causal Inference. Winter 2025
- Stats 310B/Math 230B: Theory of Probability II. Winter 2024
- Stats 310A/Math 230A: Theory of Probability I. Autumn 2023
- Stats 216: Introduction to Statistical Learning. Winter 2023
- Stats 209: Introduction to Causal Inference. Fall 2025
- Stats 202: Data Mining and Analysis. Summer 2023, Autumn 2022
- Stats 200: Introduction to Theoretical Statistics. Winter 2026, Autumn 2024
- Stats 141: Introduction to Statistics for Biology. Spring 2026
- Stats 60: Introduction to Statistical Methods: Precalculus. Spring 2025

Other experiences

- Trained numerous high school students for mathematics olympiads and other math competitions; materials are available at ghoshadi.wordpress.com [↗](#).

ACCOLADES	• Departmental Teaching Assistant Award , Stanford University	2026
	• IMS 2025 International Conference on Statistics and Data Science (ICSDS) , student travel award	2025
	• IISA conference 2025 , best poster award	2025
	• Industrial Affiliates Annual Conference at Stanford , travel award	2024
	• Selected for Machine Learning in Economics Summer Institute at University of Chicago Booth school of business	2024
	• ISIAA – J. K. Ghosh Memorial Gold Medal (for outstanding performance in Masters of Statistics)	2023
	• ISIAA – Mrs. M. R. Iyer Memorial Gold Medal (for best overall performance in Bachelor of Statistics)	2021
	• Nikhilesh Bhattacharyya Memorial Gold Medal (for best performance in major (Statistics) in Bachelor of Statistics)	2021
	• Madhava Mathematics Competition , selected among the top few participants in the country for a prestigious summer camp	2019, 2018
	• Indian National Mathematical Olympiad , earned a prestigious certificate of merit from NBHM, Govt. of India (awarded to the top 75 participants in the country)	2016
INVITED TALKS	• Metrics-lunch (an weekly seminar on Econometrics), Stanford University	2026
	Title: Non-parametric Causal Inference in Dynamic Thresholding Designs	
	• Data-driven Seminar , Stanford University	2026
	Title: Robustness and Efficiency of Rosenbaum's Rank-based Estimator in Randomized Trials: A Design-based Perspective	
	• University of Washington Causal Inference & Missing Data Reading Group	2026
	Title: Assumption-robust Causal Inference	
	• IMS International Conference on Statistics and Data Science , Seville, Spain	2025
	Session: Student Award Session 3	
	Title: Assumption-robust Causal Inference	

- **Stanford Causal Science Center (SC²) Conference**, Stanford University 2025
Title: Causal Inference in Dynamic Thresholding Designs
- **Joint Statistical Meeting**, Nashville, Tennessee 2025
Session: Regression Discontinuity Designs with Complex Data
Title: Causal Inference in Dynamic Thresholding Designs
- **Industrial Affiliates Annual Conference**, Stanford University 2024
Title: Practical bias-aware inference in regression discontinuity designs: An asymptotic view
- **Stanford Causal Science Center (SC²) Conference**, Stanford University 2024
Title: Asymptotic bias-aware inference in regression discontinuity designs under higher-order smoothness
- **Computational and Methodological Statistics**, Berlin, Germany 2023
Title: Efficiency and robustness of Rosenbaum's regression (un)-adjusted rank-based estimator in randomized experiments
- **PCM Memorial Lecture**, **Indian Statistical Institute**, Kolkata 2022
Title: The synthetic control method in causal inference
- **D. Basu Memorial Lecture**, **Indian Statistical Institute**, Kolkata 2021
Title: Large low-rank matrix completion
- **Online Reading Group on Functional Data Analysis** ↗ 2021
Title: Two-sample testing of the equality of mean functions
- **Students' Learning Seminar**, **Indian Statistical Institute**, Kolkata 2021
Title: Matching estimators in causal inference

CLASS
PROJECTS

- **SMARTer Multi-task Fine-tuning of BERT**
Summary: Engineered a multi-task NLP framework, integrating adversarial regularization, proximal optimization, and low-rank adaptation with randomized task scheduling for stable cross-task training. Joint with Disha Ghandwani and Rahul Kanekar.
- **Inference for Data Adaptively Sampled via REINFORCE**
Summary: Developed inference methods for adaptively collected data and analyzed the robustness–efficiency trade-offs in variance estimation for long-horizon policy evaluation. Joint with Ivy Zhang.
- **Predicting the Number of Micro-businesses Across the U.S.**
Summary: Built a forecasting pipeline using GoDaddy panel and Census data, applying ARIMA, GLMs, and varying-coefficient models to link demographics with county-level microbusiness dynamics. Joint with Joshua Kazdan and Yash Nair.

SKILLS

Python, **R**, **Markdown**, **Git**, **C** (basic), **HTML** (basic). **R package**: plrd ↗